Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Raizia mam
Student Name	Keerthi Garikipati
Roll Number	2520030416
Branch	CSE
Course Outcome	CO-2

Case Study Topic: healthnet - intelligent hospital management system

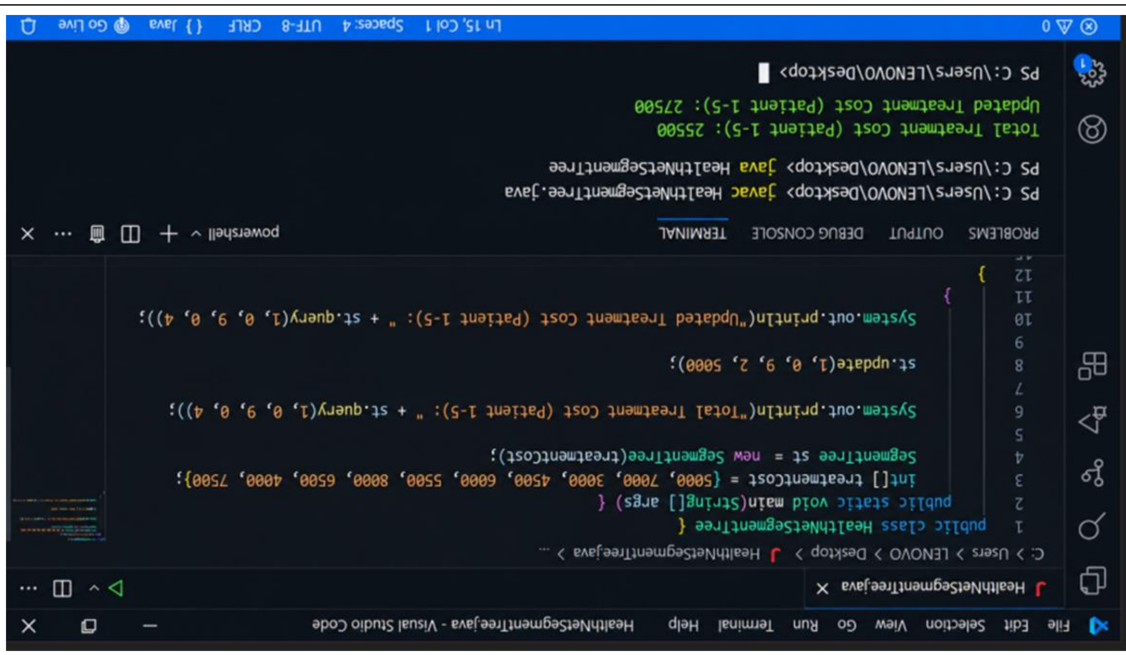
Work Done Summary:

In this case study, we implemented a Segment Tree to manage and analyze hospital patient data efficiently. The Segment Tree stores patient treatment costs and supports fast range queries and updates. The system allows hospital administrators to quickly calculate the total treatment cost for a group of patients and update patient bills when necessary.
We inserted 10 patient treatment records and constructed a Segment Tree. The tree was traversed to process range sum queries and point updates efficiently. We also compared Segment Tree performance against traditional array traversal methods for large hospital datasets.

Implementation details:

1. Inserted 10 patient treatment cost records into an array.
2. Constructed a Segment Tree from the treatment cost data.
3. Implemented range sum query to calculate total treatment costs within a specified range.
4. Implemented point update operation to modify treatment costs.
5. Reduced query complexity from O(n) to O(log n).
6. Used recursive tree construction and traversal techniques.
7. Compared Segment Tree performance with normal array traversal.
8. Improved efficiency for large-scale hospital databases.

Result:



Github link:

Student signature	Faculty signature